17B 1 DIFFERENT TYPES OF REACTIONS

SPECIFICATION REFERENCE
17.11 17.24(i)
17.24(ii) CP14

LEARNING OBJECTIVES

- Understand that colour changes in transition metal ions may arise as a result of changes in:
  - (i) oxidation number of the ion
  - (ii) ligand
  - (iii) coordination number of the complex.
- Understand that ligand exchange, and an accompanying colour change, occurs in the formation of:
  - (i) $[Cu(NH_3)_4(H_2O)_2]^{2+}$ from $[Cu(H_2O)_6]^{2+}$ via $Cu(OH)_2(H_2O)_4$
  - (ii) $[CuCl_4]^{2-}$ from $[Cu(H_2O)_6]^{2+}$.
- Be able to write ionic equations to show the meaning of amphoteric behaviour, deprotonation and ligand exchange reactions.

TYPES OF REACTIONS

So far, we have considered the origin of colour in transition metal ions. We can now consider why there are often colour changes when transition metal ions take part in reactions. Four main types of reactions can occur:

- redox – the oxidation number of the transition metal ion changes
- **deprotonation** – one or more of the ligands gains or loses a hydrogen ion (proton)
- **ligand exchange** – one or more of the ligands around the transition metal ion is replaced by a different ligand
- coordination number change – the number of ligands changes.

Any one of these types of reactions can cause a change in the colour of the complex. Some reactions involve more than one of these types of reactions.

CHANGE IN OXIDATION NUMBER

An aqueous solution containing $Fe^{2+}(aq)$ ions is pale green, but when it is exposed to air it gradually turns yellow or brown, as the oxidation number of iron increases from +2 to +3. The type and number of ligands remain unchanged in this oxidation reaction, so the formulae of the two complexes are $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$. Colour changes such as the one in this reaction are best illustrated using solid samples containing the ions (see **fig A**).

Equations are not usually written for oxidation reactions in which the only change is the oxidation number of the transition metal ion.



Iron(II) sulphate ($FeSO^4$) Iron(III) sulphate ($Fe2(SO^4)^3$)

▲ **fig A** Solid samples clearly show colour differences between ions.

FORMATION OF $[Cu(NH_3)_4(H_2O)_2]^{2+}$: DEPROTONATION AND LIGAND EXCHANGE REACTIONS

Consider the reaction that occurs when aqueous sodium hydroxide is added to copper(II) sulfate solution. The observation is that a pale blue solution forms a blue precipitate. The equation for this reaction is:

$$[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow [Cu(H_2O)_4(OH)_2] + 2H_2O$$

You might think that this is a ligand substitution reaction – that two hydroxide ions have replaced two water molecules. In fact, it is a deprotonation reaction – the two hydroxide ions have removed hydrogen ions from two of the water ligands and converted them into water molecules. The two water ligands that have lost hydrogen ions are now hydroxide ligands.

PRACTICAL SKILLS CP14

Tetramminecopper(II) sulfate-1-water, $[Cu(NH_3)_4.SO_4].H_2O$, can be prepared by adding aqueous ammonia to an aqueous solution of copper(II) sulfate.

The overall equation for the reaction is:

$$[Cu(H_2O)_6]^{2+} + 4NH_3 + SO_4^{2-} \rightarrow [Cu(NH_3)_4].SO_4.H_2O + 5H_2O$$

This is an example of a ligand exchange reaction. This is the reaction you might investigate in **CP14: The preparation of a transition metal complex**.

Exactly the same observations can be made during the careful addition of aqueous ammonia instead of aqueous sodium hydroxide. The equation for this reaction is:

$$[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(H_2O)_4(OH)_2] + 2NH_4^+$$

DID YOU KNOW?

The formula for the copper(II)-ammine complex in aqueous solution is sometimes given as $[Cu(NH_3)_4]^{2+}$. This is *not* correct. The correct formula is shown in the text. The confusion arose because the bonds from the Cu^{2+} ion to the water ligands are longer than the bonds from the Cu^{2+} ion to the ammonia ligands. This is the result of something called the Jahn–Teller effect. The explanation for this effect is beyond the aims of this book.



fig B The pale blue solution contains $[Cu(H_2O)_6]^{2+}$, the pale blue precipitate contains $[Cu(H_2O)_4(OH)_2]$ and the deep blue solution contains $[Cu(NH_3)_4(H_2O)_2]^{2+}$.



fig C The pale blue solution contains $[Cu(H_2O)_6]^{2+}$, the yellow solution on the right contains $[CuCl_4]^{2-}$ and the green solution in the middle contains a mixture of the two.

SUBJECT VOCABULARY

deprotonation the removal of one or more hydrogen ions (protons) from a complex ion
ligand exchange when one ligand in a complex ion is replaced by a different ligand
amphoteric (substance) a substance that can act both as an acid and as a base
amphoteric behaviour the ability of a species to react with both acids and bases

It is again easy to see that this equation involves a deprotonation – two of the water ligands transfer a hydrogen ion to the ammonia molecules.

When aqueous ammonia is added to the blue precipitate formed, it dissolves to form a deep blue solution. The equation for this reaction is:

$$[Cu(H_2O)_4(OH)_2] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2H_2O + 2OH^-$$

This is a ligand exchange reaction – four ammonia molecules replace two water molecules and two hydroxide ions. The solutions at the start and end of the reaction and the intermediate precipitate are shown in **fig B**.

FORMATION OF $[CuCl_4]^{2-}$: CHANGE IN COORDINATION NUMBER

These reactions always involve a change of ligand as well as a change in coordination number. A good example is the reaction between copper(II) sulfate solution and concentrated hydrochloric acid. When the acid is added slowly and continuously, the colour gradually changes from blue to green and finally to yellow. This is the equation for the reaction:

$$[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$$

The state symbols, all (aq), have been left out for clarity.

You can see that all six water ligands have been replaced by four chloride ions. This reaction is also an example of a change in coordination number, from 6 to 4. Note that although the charge on the complex has changed from 2+ to 2−, there has been no change in oxidation number.

The arrow shows that the reaction is reversible, which helps to explain the colour change observed. The hexaaquacopper(II) ion is blue and the tetrachlorocuprate(II) ion is yellow, so the green colour is due to a mixture of the blue and yellow complex ions. The solutions at the start and end of the reaction and the intermediate mixture are shown in **fig C**.

LEARNING TIP

You should remember the colours of the species in these reactions, but you do not need to understand why the species have the actual colours.

AMPHOTERIC HYDROXIDES

We have already seen that hydrated transition metal ions can be deprotonated by adding a base such as aqueous sodium hydroxide to form a precipitate of the metal hydroxide. For example, chromium(III) hydroxide is precipitated from an aqueous solution containing Cr^{3+} ions:

$$[Cr(H_2O)_6]^{3+}(aq) + 3OH^-(aq) \rightarrow [Cr(H_2O)_3(OH)_3](s) + 3H_2O(l)$$
green solution — green precipitate

When an excess of sodium hydroxide is added to this precipitate, further deprotonation can take place:

$$[Cr(H_2O)_3(OH)_3](s) + 3OH^-(aq) \rightarrow [Cr(OH)_6]^{3-}(aq) + 3H_2O(l)$$
green solution

In this reaction, chromium(III) hydroxide is acting as an acid, since it is reacting with a base.

Chromium(III) hydroxide is also a base, because it can react with acids:

$$[Cr(H_2O)_3(OH)_3](s) + 3H^+(aq) \rightarrow [Cr(H_2O)_6]^{3+}(aq)$$

A metal hydroxide that can act as both an acid and a base is called an **amphoteric** hydroxide. This is an example of **amphoteric behaviour**.

CHECKPOINT

1. The equation for a reaction of a transition metal ion is:
   $$[Fe(H_2O)_6]^{3+} + SCN^- \rightarrow [Fe(SCN)(H_2O)_5]^{2+} + H_2O$$
   Explain which of the four main types of reactions are illustrated by this equation.
2. The complex $[Co(H_2O)_6]^{2+}$ is converted into $[Co(NH_3)_6]^{3+}$.
   Explain which types of reactions occur in this conversion.

17B 2 REACTIONS OF COBALT AND IRON COMPLEXES

SPECIFICATION REFERENCE
17.22 PART 17.24(iii)

LEARNING OBJECTIVES

- Be able to record observations and write suitable equations for the reactions of $Fe^{2+}(aq)$, $Fe^{3+}(aq)$ and $Co^{2+}(aq)$ with aqueous sodium hydroxide and aqueous ammonia, including in excess.
- Understand that ligand exchange, and an accompanying colour change, occurs in the formation of $[CoCl_4]^{2-}$ from $[Co(H_2O)_6]^{2+}$.

REACTIONS INVOLVING COBALT COMPLEXES



▲ **fig A** The bright colour of these glass bottles is caused by blue cobalt compounds.

REACTION WITH ALKALIS

Consider the reaction that occurs when aqueous sodium hydroxide is added to a solution containing the hexaaquacobalt(II) ion until no further change is seen. The observation is that a pink solution forms a blue precipitate. The equation for this reaction is:

$$[Co(H_2O)_6]^{2+} + 2OH^- \rightarrow [Co(H_2O)_4(OH)_2] + 2H_2O$$

As with copper, this is a deprotonation reaction – the two hydroxide ions have removed hydrogen ions from two of the water ligands and converted them into water molecules. The two water ligands that have lost hydrogen ions are now hydroxide ligands. Upon standing, the colour of the precipitate gradually changes to pink.

The same observations can be made when aqueous ammonia is used as the alkali. However, when aqueous ammonia is added to excess, there is an observation not made with aqueous sodium hydroxide – the precipitate dissolves to form a pale yellow solution. The equation for the deprotonation reaction is:

$$[Co(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Co(H_2O)_4(OH)_2] + 2NH_4^+$$

The equation for the ligand exchange reaction forming the pale yellow solution is:

$$[Co(H_2O)_4(OH)_2] + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 4H_2O + 2OH^-$$

Some features of transition metal chemistry are impossible to explain without going well beyond the aims of this book. One of these is why six ammonia ligands are involved in the reaction with the cobalt complex, while in the corresponding reaction with copper only four ammonia ligands are involved.

Upon standing, this pale yellow solution changes colour because of oxidation by the oxygen in the atmosphere. The oxidation number of cobalt increases from +2 to +3, and the darker yellow $[Co(NH_3)_6]^{3+}$ ion forms. Unfortunately, the resulting solution usually looks brown rather than yellow. This is because other products are formed, from ligand exchange reactions with both water molecules and any negative ions present.



fig B Ordinary glass is often very pale green. This is more noticeable if you look through a considerable thickness of glass. The green colour is caused by the presence of impurities containing iron.

REACTION WITH CONCENTRATED HYDROCHLORIC ACID

This reaction is very like the one with the $[Cu(H_2O)_6]^{2+}$ ion. When concentrated hydrochloric acid is slowly added to a solution containing the hexaaquacobalt(II) ion, the pink solution gradually changes to blue. This is the equation for the reaction:

$$[Co(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CoCl_4]^{2-} + 6H_2O$$

You can see that all six water ligands have been replaced by four chloride ions. This reaction is also an example of a change in coordination number, from 6 to 4. Note that although the charge on the complex has changed from 2+ to 2−, there has been no change in oxidation number.

REACTIONS INVOLVING IRON COMPLEXES

REACTION OF IRON(II) COMPLEXES WITH ALKALIS

Consider the reaction that occurs when aqueous sodium hydroxide is added to a solution containing the hexaaquairon(II) ion until no further change is seen. The observation is that a pale green solution forms a green precipitate. The equation for this reaction is:

$$[Fe(H_2O)_6]^{2+} + 2OH^- \rightarrow [Fe(H_2O)_4(OH)_2] + 2H_2O$$

As with copper and cobalt, this is a deprotonation reaction – the two hydroxide ions have removed hydrogen ions from two of the water ligands and converted them into water molecules. The two water ligands that have lost hydrogen ions are now hydroxide ligands.

The same observations can be made when aqueous ammonia is used as the alkali. The equation for the deprotonation reaction is:

$$[Fe(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Fe(H_2O)_4(OH)_2] + 2NH_4^+$$

Upon standing, the colour of the green precipitate gradually changes to brown as oxygen from the atmosphere causes oxidation, forming $[Fe(H_2O)_3(OH)_3]$ – this is the triaquatrihydroxoiron(III) complex.

REACTION OF IRON(III) COMPLEXES WITH ALKALIS

Consider the reaction that occurs when aqueous sodium hydroxide is added to a solution containing the hexaaquairon(III) ion until no further change is seen. The observation is that a yellow-brown solution forms a brown precipitate. The equation for this reaction is:

$$[Fe(H_2O)_6]^{3+} + 3OH^- \rightarrow [Fe(H_2O)_3(OH)_3] + 3H_2O$$

As with iron(II), this is a deprotonation reaction – the three hydroxide ions have removed hydrogen ions from three of the water ligands and converted them into water molecules. The three water ligands that have lost hydrogen ions are now hydroxide ligands.

The same observations can be made when aqueous ammonia is used as the alkali. The equation for the deprotonation reaction is:

$$[Fe(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Fe(H_2O)_3(OH)_3] + 3NH_4^+$$

There are no further reactions when an excess of either aqueous sodium hydroxide or aqueous ammonia is added and no further changes upon standing.

EXAM HINT

You will always be asked questions on transition metal complexes in your exam. To answer these questions well you must learn the reactions, remember the colours and make sure you use the correct notation.

LEARNING TIP

Note that atmospheric oxidation occurs with complexes containing a transition metal ion with a +2 oxidation number, but not for those with a +3 oxidation number.

CHECKPOINT

1. When aqueous ammonia is added in excess to a solution containing $[Co(H_2O)_6]^{2+}$ ions, an acid–base reaction occurs, followed by a ligand exchange reaction. Write an equation to describe this reaction.

SKILLS REASONING/ ARGUMENTATION

2. A student wrote this equation to explain the formation of the brown precipitate when aqueous ammonia was added to a solution containing Fe^{3+} ions:

   $$[Fe(H_2O)_6]^{3+} + 3OH^- \rightarrow [Fe(H_2O)_3(OH)_3] + 3H_2O$$

   Why is the above reaction less likely to occur than the one shown below?

   $$[Fe(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Fe(H_2O)_3(OH)_3] + 3NH_4^+$$

17B 3 THE CHEMISTRY OF CHROMIUM

SPECIFICATION REFERENCE
17.20(i) 17.20(ii) 17.21 17.22 PART

LEARNING OBJECTIVES

- Be able to record observations and write suitable equations for the reactions of $Cr^{3+}(aq)$ with aqueous sodium hydroxide and aqueous ammonia, including in excess.
- Understand, in terms of the relevant *E* values, that the dichromate(VI) ion, $Cr_2O_7^{2-}$:
  (i) can be reduced to Cr^{3+} and Cr^{2+} ions using zinc in acidic conditions
  (ii) can be produced by the oxidation of Cr^{3+} ions using hydrogen peroxide in alkaline conditions (followed by acidification).
- Know that the dichromate(VI) ion, $Cr_2O_7^{2-}$, can be converted into chromate(VI) ions as a result of the equilibrium
  $Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$

INTRODUCTION

Some of the reactions of chromium are similar to those we have already met for the ions of copper, cobalt, iron(II) and iron(III). However, chromium has ions in which the metal has an oxidation number of +6, so reactions involving these ions must be considered in a different way.

A WORD OF WARNING

We have already discussed that there are an infinite number of colours in the spectrum of visible light, although we sometimes refer to the seven colours of the rainbow. The perception of colours by humans varies a lot. A specific colour might be described as pink or purple by different people, and a colour described as blue-green by one person might be described as blue, green or turquoise by others.

The situation with the colours used to describe chromium compounds is further complicated by these factors:

- Some compounds have different colours as solids and aqueous solutions.
- The colour of a solution depends on concentration.
- The presence of dissolved oxygen in an aqueous solution can affect the colour observed.

This means that the colours used in this book may not always correspond to those in other books, and may be different from those observed in test tube reactions that you see.

REACTIONS INVOLVING CHROMIUM(III) COMPLEXES

REACTION OF CHROMIUM(III) COMPLEXES WITH ALKALIS

Consider the reaction that occurs when aqueous sodium hydroxide is added to a solution containing the hexaaquachromium(III) ion until no further change is seen. The observation is that a green solution forms a green precipitate. The equation for this reaction is:

$$[Cr(H_2O)_6]^{3+} + 3OH^- \rightarrow [Cr(H_2O)_3(OH)_3] + 3H_2O$$

LEARNING TIP

The colour of the $[Cr(NH_3)_6]^{3+}$ ion in a solid, such as chrome alum, is violet. However, when this solid is dissolved in water the resulting solution is green, not violet. This is because of ligand exchange reactions that occur with the negative ions present.

As with iron(III), this is a deprotonation reaction – the three hydroxide ions have removed hydrogen ions from three of the water ligands and converted them into water molecules. The three water ligands that have lost hydrogen ions are now hydroxide ligands.

The same observations can be made when aqueous ammonia is used as the alkali. The equation for the deprotonation reaction is:

$$[Cr(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Cr(H_2O)_3(OH)_3] + 3NH_4^+$$

EXAM HINT

In reactions with transition metal complexes, make sure you know *when* and *how* to show ammonia acting as a base and ammonia acting as a ligand.

When an excess of aqueous sodium hydroxide is added, the green precipitate dissolves to form a green solution. The equation for this deprotonation reaction can be represented as:

$$[Cr(H_2O)_3(OH)_3] + OH^- \rightarrow [Cr(H_2O)_2(OH)_4]^- + H_2O$$

If the aqueous sodium hydroxide is more concentrated, further deprotonation reactions occur, such as

$$[Cr(H_2O)_2(OH)_4]^- + 2OH^- \rightarrow [Cr(OH)_6]^{3-} + 2H_2O$$

although there is no further change in colour.

The reactions involving hydroxide ions can be reversed by the addition of acid, illustrating the amphoteric nature of the neutral complex.

When an excess of aqueous ammonia is added to the green precipitate, the precipitate is slow to dissolve, but eventually a violet or purple solution forms:

$$[Cr(H_2O)_3(OH)_3] + 6NH_3 \rightarrow [Cr(NH_3)_6]^{3+} + 3H_2O + 3OH^-$$

In all of the reactions so far, the oxidation number of chromium has remained unchanged at +3. However, provided that the solutions are alkaline, oxidation is easily achieved by the addition of the oxidising agent hydrogen peroxide. In this reaction the solution changes from green to yellow as the chromate(VI) ion, with oxidation number +6, is formed. The equation for this oxidation is:

$$2[Cr(OH)_6]^{3-} + 3H_2O_2 \rightarrow 2CrO_4^{2-} + 2OH^- + 8H_2O$$

Note that although the chromate(VI) ion is a complex, it is not enclosed in square brackets.

CHROMATE(VI) AND DICHROMATE(VI) IONS

Chromate(VI) ions are stable in alkaline solution, but in acidic conditions the dichromate(VI) ion is more stable. So, if acid is added there is a colour change from yellow to orange as the following reaction occurs:

$$2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$$

This reaction is easily reversed by adding alkali. When considering redox reactions, it is often easier to use simplified formulae by leaving out square brackets and ligands that do not undergo redox reactions, especially water.

REDUCTION OF DICHROMATE(VI) IONS

When zinc metal is added to an acidic solution containing dichromate(VI) ions, reduction reactions occur in which the oxidation number of chromium decreases first to +3 and then to +2.

The first stage of the reduction involves a colour change from orange to green, as this reaction occurs:

$$Cr_2O_7^{2-} + 14H^+ + 3Zn \rightarrow 2Cr^{3+} + 7H_2O + 3Zn^{2+}$$

The second stage of the reduction involves a colour change from green to blue, as this reaction occurs:

$$2Cr^{3+} + Zn \rightarrow 2Cr^{2+} + Zn^{2+}$$

EXPLANATION OF REDOX REACTIONS USING $E^\ominus$ VALUES

It is often helpful to explain why redox reactions occur by considering the standard electrode potentials of the different redox systems involved. For chromium, we need to consider these values:

1	$Zn^{2+} + 2e^- \rightleftharpoons Zn$	$E^\ominus = -0.76\ V$
2	$Cr^{3+} + e^- \rightleftharpoons Cr^{2+}$	$E^\ominus = -0.41\ V$
3	$CrO_4^{2-} + 4H_2O + 3e^- \rightleftharpoons Cr(OH)_3 + 5OH^-$	$E^\ominus = -0.13\ V$
4	$H_2O_2 + 2e^- \rightleftharpoons 2OH^-$	$E^\ominus = +1.24\ V$
5	$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	$E^\ominus = +1.33\ V$

If you have already learned about standard electrode potentials earlier in this book you may remember that when half-equations are arranged from high negative $E^\ominus$ values at the top to high positive values at the bottom, then the best reducing agent is at the top and on the right, and the best oxidising agent is at the bottom and on the left.

EXPLAINING OXIDATION FROM +3 TO +6

The simplified equation for the oxidation of chromium(III) by hydrogen peroxide in alkaline conditions is:

$$2Cr(OH)_3 + 3H_2O_2 + 4OH^- \rightarrow 2CrO_4^{2-} + 8H_2O$$

How can this equation be obtained from the half-equations?

The relevant half-equations are 3 and 4 (see previous page). In these equations, the best reducing agent is $Cr(OH)_3$, and the best oxidising agent is H_2O_2. For these two species to react together, we need to add half-equation 3 (reversed) to half-equation 4. When adding half-equations, they may need to be multiplied so that the number of electrons is the same in both. In this example, we need to multiply equation 3 by 2, and equation 4 by 3, so as to obtain $6e^-$ in both:

$$2Cr(OH)_3 + 10OH^- \rightleftharpoons 2CrO_4^{2-} + 8H_2O + 6e^-$$
$$3H_2O_2 + 6e^- \rightleftharpoons 6OH^-$$

Adding the half-equations and cancelling identical species on both sides gives:

$$2Cr(OH)_3 + 3H_2O_2 + 4OH^- \rightarrow 2CrO_4^{2-} + 8H_2O$$

Since the $E^\ominus$ value for half-equation 3 is more negative than the $E^\ominus$ value for half-equation 4, $Cr(OH)_3$ is electron releasing with respect to H_2O_2, and therefore the reaction is thermodynamically feasible. If the solution is then acidified by adding dilute sulfuric acid, the chromate(VI) ion is converted into the dichromate(VI) ion:

$$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$$

EXPLAINING REDUCTION FROM +3 TO +6

The simplified equation for the reduction of chromium(VI) by zinc in acidic conditions is:

$$Cr_2O_7^{2-} + 14H^+ + 3Zn \rightarrow 2Cr^{3+} + 7H_2O + 3Zn^{2+}$$

How can this equation be obtained from the half-equations?

The relevant half-equations are 1 and 5 (see previous page). In these equations, the best reducing agent is Zn and the best oxidising agent is $Cr_2O_7^{2-}$. For these two species to react together, we need to add half-equation 1 (reversed) to half-equation 6, bearing in mind multiplying half-equations:

$$3Zn \rightleftharpoons 3Zn^{2+} + 6e^-$$
$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$$

As in the previous example, adding gives:

$$Cr_2O_7^{2-} + 14H^+ + 3Zn \rightarrow 2Cr^{3+} + 7H_2O + 3Zn^{2+}$$

Since the $E^\ominus$ value for half-equation 1 is more negative than the $E^\ominus$ value for half-equation 5, Zn is electron releasing with respect to $Cr_2O_7^{2-}$, and therefore the reaction is thermodynamically feasible.

EXPLAINING REDUCTION FROM +3 TO +2

The simplified equation for the reduction of chromium(III) by zinc in acidic conditions is:

$$2Cr^{3+} + Zn \rightarrow 2Cr^{2+} + Zn^{2+}$$

How can this equation be obtained from the half-equations?

The relevant half-equations are 1 and 2. In these equations, the best reducing agent is Zn and the best oxidising agent is Cr^{3+}. For these two species to react together, we need to add half-equation 1 (reversed) to half-equation 2, bearing in mind multiplying half-equations:

$$Zn \rightleftharpoons Zn^{2+} + 2e^-$$
$$2Cr^{3+} + 2e^- \rightleftharpoons 2Cr^{2+}$$

As in the previous example, adding gives:

$$Cr^{3+} + Zn \rightarrow 2Cr^{2+} + Zn^{2+}$$

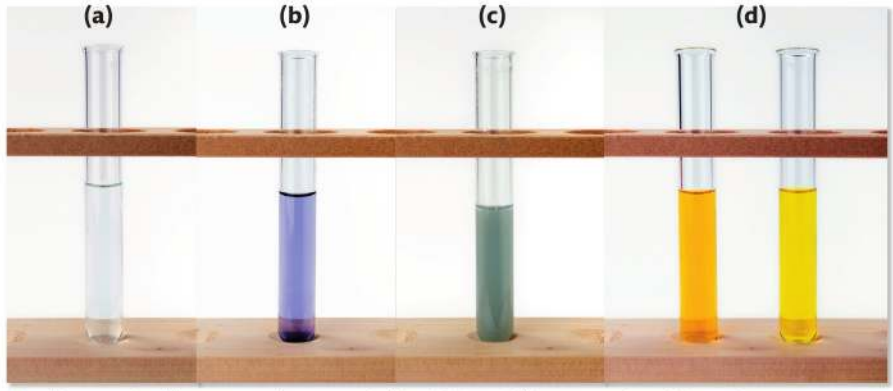
Since the $E^\ominus$ value for half-equation 1 is more negative than the $E^\ominus$ value for half-equation 2, Zn is electron releasing with respect to Cr^{3+}, and therefore the reaction is thermodynamically feasible.

CHROMIUM CHEMISTRY SUMMARY

Table A summarises the important reactions of chromium complexes in four sequences. Some of the coloured solutions are shown in **fig A**.

START	REAGENT	INTERMEDIATE	REAGENT	FINISH
$[Cr(H_2O)_6]^{3+}$ green solution	add NaOH(aq)	$[Cr(H_2O)_3(OH)_3]$ green precipitate	add NaOH(aq)	$[Cr(OH)_6]^{3-}$ green solution
$[Cr(H_2O)_6]^{3+}$ green solution	add NH_3(aq)	$[Cr(H_2O)_3(OH)_3]$ green precipitate	add NH_3(aq)	$[Cr(NH_3)_6]^{3+}$ violet solution
$[Cr(OH)_6]^{3-}$ green solution	add H_2O_2/OH^-	CrO_4^{2-} yellow solution	add H^+	$Cr_2O_7^{2-}$ orange solution
CrO_4^{2-} yellow solution	add Zn/H^+	$[Cr(H_2O)_6]^{3+}$ green solution	add Zn/H^+	$[Cr(H_2O)_6]^{2+}$ blue solution

table A



▲ **fig A** Tube (a) contains the very pale blue $[Cr(H_2O)_6]^{2+}$ ion, produced by reduction using zinc. Tube (b) contains the $[Cr(H_2O)_6]^{3+}$ ion, which is normally green, but can appear violet in the presence of other ions, as here. Tube (c) contains the $[Cr(OH)_6]^{3-}$ ion, which is also green, present in alkaline solutions. The tubes in (d) contain chromium with oxidation number +6. The orange solution contains the $Cr_2O_7^{2-}$ ion and the yellow solution contains the CrO_4^{2-} ion.

LEARNING TIP

Understanding chromium chemistry involves more effort than for some other transition metals because of the number of different ions. As well as all of the complexes involving Cr^{2+} and Cr^{3+}, there are two complexes with oxidation number +6 (CrO_4^{2-} and $Cr_2O_7^{2-}$).

CHECKPOINT

SKILLS REASONING/ARGUMENTATION

1. Excess acid is added to a green solution containing $[Cr(OH)_6]^{3-}$ ions. Write an equation for the reaction in which a different green chromium-containing solution is formed.
2. The standard electrode potential for the redox system $Mn^{2+} + 2e^- \rightleftharpoons Mn$ is −1.18 V.
   Explain whether or not zinc can reduce Mn^{2+} ions to the element Mn.

17B 4 REACTIONS OF MANGANESE COMPLEXES

SPECIFICATION REFERENCE
17.22 PART 17.23 PART

LEARNING OBJECTIVES

- Be able to record observations and write suitable equations for the reactions of $Mn^{2+}(aq)$ with aqueous sodium hydroxide and aqueous ammonia, including in excess.
- Be able to write ionic equations to show the meaning of deprotonation and ligand exchange in the reactions of manganese ions.

REACTIONS OF MANGANESE(II) COMPLEXES WITH ALKALIS

Consider the reaction that occurs when aqueous sodium hydroxide is added to a solution containing the hexaaquamanganese(II) ion until no further change is seen. The observation is that a pale pink solution forms a pale brown precipitate. The equation for the reaction is:

$$[Mn(H_2O)_6]^{2+} + 2OH^- \rightarrow [Mn(H_2O)_4(OH)_2] + 2H_2O$$

This is a deprotonation reaction – the two hydroxide ions have removed hydrogen ions from two of the water ligands and converted them into water molecules. The two water ligands that have lost hydrogen ions are now hydroxide ligands. The pale brown precipitate turns darker brown on standing in air as it is oxidised to $[Mn(H_2O)_3(OH)_3]$, and then turns very dark brown, forming hydrated manganese(IV) oxide, $MnO_2.xH_2O$.

The precipitate does not dissolve in excess aqueous sodium hydroxide.

The same observations are made when aqueous ammonia is used as the alkali. The equation for this deprotonation reaction, forming the pale brown precipitate, is:

$$[Mn(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Mn(H_2O)_4(OH)_2] + 2NH_4^+$$

LEARNING TIP

There are two important points you need to be aware of for the reaction between manganese(II) ions and aqueous sodium hydroxide. The first is that dilute solutions containing $[Mn(H_2O)_6]^{2+}$ are very likely to appear colourless, not pale pink. The second is that the true colour of $[Mn(H_2O)_4(OH)_2]$ is white, but the white colour may not be seen as the precipitate rapidly turns pale brown.

CHECKPOINT

SKILLS REASONING/ARGUMENTATION

1. An aqueous solution of an inorganic solid, **Q**, forms a pale brown precipitate, **R**, when aqueous sodium hydroxide is added. **R** darkens on standing in air to form a dark brown solid, **S**. When solid **S** is heated it forms a dark brown metal oxide, **T**O_2.

(a) **T**O_2 contains 36.82% by mass of oxygen. Use this information to calculate the molar mass of the metal **T**, and use this to identify **T**.

(b) Identify the cation present in the aqueous solution of **Q**.

(c) Identify **R** and **S**.

2. For the reactions above:

(a) Write an ionic equation for the formation of **R** from the solution of **Q**. Include state symbols.

(b) Write an equation for the action of heat on solid **S**.

17B 5 THE CHEMISTRY OF VANADIUM

SPECIFICATION REFERENCE
17.18 17.19

LEARNING OBJECTIVES

- Know the colours of the oxidation states of vanadium (+5, +4, +3 and +2) in its compounds.
- Understand redox reactions for the interconversion of the oxidation states of vanadium (+5, +4, +3 and +2), in terms of the relevant $E^{\ominus}$ values.

EXAM HINT

You can easily lose marks in an exam question if you are careless when showing the difference between VO_2^+ and VO^{2+} in your answer. Make sure you write your answer clearly.

REDOX REACTIONS

Like chromium, vanadium is a transition metal that forms ions with several oxidation numbers. **Table A** shows the important ones.

OXIDATION NUMBER	FORMULA	NAME	COLOUR OF AQUEOUS SOLUTION
+2	V^{2+}	vanadium(II)	purple
+3	V^{3+}	vanadium(III)	green
+4	VO^{2+}	oxovanadium(IV)	blue
+5	VO_2^+	dioxovanadium(V)	yellow

table A

Unlike with chromium and the other transition elements, the focus here with vanadium is purely on redox reactions. Vanadium is a good choice for this because there is one distinct colour for each of the main oxidation numbers found in its compounds. So, it is relatively easy to demonstrate the change in oxidation number by observing the change in colour during the reaction. As with chromium, we will consider the feasibility of these redox reactions in terms of $E^{\ominus}$ values. Although most of the species involved are complexes, and some contain water ligands, the square brackets and water ligands are left out for clarity.

REDUCING VANADIUM FROM +5 TO +2

The usual source of vanadium with oxidation number +5 is the compound ammonium vanadate(V), NH_4VO_3. An acidic solution of this compound contains the dioxovanadium(V) ion, VO_2^+. When zinc is added to this solution, reduction begins and there is a gradual colour change from yellow, through blue and green, to purple as the oxidation number decreases from +5 to +2. All of these colours are of solutions – there are no precipitates involved. The test tubes in **fig A** show the colours.



▲ **fig A** The tube on the left (yellow) contains the VO_2^+ ion. The second tube (blue) contains the VO^{2+} ion. The third tube (green) contains the V^{3+} ion. The tube on the right (purple) contains the V^{2+} ion.

EXPLANATION USING $E^{\ominus}$ VALUES

For vanadium, we need to consider these values:

1	$V^{2+} + 2e^- \rightleftharpoons V$	$E^{\ominus} = -1.18\,V$
2	$Zn^{2+} + 2e^- \rightleftharpoons Zn$	$E^{\ominus} = -0.76\,V$
3	$V^{3+} + e^- \rightleftharpoons V^{2+}$	$E^{\ominus} = -0.26\,V$
4	$VO_2^+ + 2H^+ + e^- \rightleftharpoons V^{3+} + H_2O$	$E^{\ominus} = +0.34\,V$
5	$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$	$E^{\ominus} = +1.00\,V$

If you followed the explanations for the redox reactions of chromium in the previous topic, you should be able to understand a shortened explanation for each stage that just shows the two relevant half-equations and then the overall equation.

REDUCTION FROM +5 TO +4

$$VO_2^+ + 2H^+ + e^- \rightarrow VO^{2+} + H_2O$$
$$Zn \rightarrow Zn^{2+} + 2e^-$$

Overall reaction:

$$2VO_2^+ + 4H^+ + Zn \rightarrow 2VO^{2+} + Zn^{2+} + 2H_2O$$

Since the $E^{\ominus}$ value for half-equation 2 is more negative than the $E^{\ominus}$ value for half-equation 5, Zn is electron releasing with respect to VO_2^+. Therefore, the reaction is thermodynamically feasible.

REDUCTION FROM +4 TO +3

$$VO^{2+} + 2H^+ + e^- \rightarrow V^{3+} + H_2O$$
$$Zn \rightarrow Zn^{2+} + 2e^-$$

Overall reaction:

$$2VO^{2+} + 4H^+ + Zn \rightarrow 2V^{3+} + Zn^{2+} + 2H_2O$$

Since the $E^{\ominus}$ value for half-equation 2 is more negative than the $E^{\ominus}$ value for half-equation 4, Zn is electron releasing with respect to VO^{2+}. Therefore, the reaction is thermodynamically feasible.

REDUCTION FROM +3 TO +2

$$V^{3+} + e^- \rightarrow V^{2+}$$
$$Zn \rightarrow Zn^{2+} + 2e^-$$

Overall reaction:

$$2V^{3+} + Zn \rightarrow 2V^{2+} + Zn^{2+}$$

Since the $E^{\ominus}$ value for half-equation 2 is more negative than the $E^{\ominus}$ value for half-equation 3, Zn is electron releasing with respect to V^{3+}. Therefore, the reaction is thermodynamically feasible.

REDUCTION FROM +2 TO 0

$$V^{2+} + 2e^- \rightarrow V$$
$$Zn \rightarrow Zn^{2+} + 2e^-$$

Overall reaction:

$$2V^{2+} + Zn \rightarrow 2V + Zn^{2+}$$

Since the $E^{\ominus}$ value for half-equation 2 is less negative than the $E^{\ominus}$ value for half-equation 1, Zn is not electron releasing with respect to V^{2+}. Therefore, the reaction is not thermodynamically feasible.

PREDICTING OXIDATION REACTIONS

A similar method can be used to predict whether a given oxidising agent will oxidise a vanadium species to one with a higher oxidation number. You can see how to do this in Question 2 in this topic.

LEARNING TIP

It is important that you do not get confused between the two oxo ions of vanadium: VO_2^+ and VO^{2+}.

CHECKPOINT

SKILLS PROBLEM-SOLVING

1. The standard electrode potential for the redox system $Sn^{2+} + 2e^- \rightleftharpoons Sn$ is −0.14 V.

Explain why the use of tin as a reducing agent will not obtain V^{2+} from a solution containing VO_2^+ ions.

2. The standard electrode potential for the redox system $Cu^{2+} + 2e^- \rightleftharpoons Cu$ is +0.34 V.

Explain whether copper(II) ions can be used to oxidise VO^{2+} ions to VO_2^+ ions.

17B 6 REACTIONS OF NICKEL AND ZINC COMPLEXES

SPECIFICATION REFERENCE
17.22 PART 17.23

LEARNING OBJECTIVES

- Be able to record observations and write suitable equations for the reactions of $Ni^{2+}(aq)$ and $Zn^{2+}(aq)$ with aqueous sodium hydroxide and aqueous ammonia, including in excess.
- Be able to write ionic equations to show the meaning of amphoteric behaviour, deprotonation and ligand exchange in the reactions of nickel ions and zinc ions.

REACTIONS OF NICKEL COMPLEXES WITH ALKALIS

Consider the reaction that occurs when aqueous sodium hydroxide is added to a solution containing the hexaaquanickel(II) ion until no further change is seen. The observation is that a green solution forms a green precipitate. The equation for the reaction is:

$$[Ni(H_2O)_6]^{2+} + 2OH^- \rightarrow [Ni(H_2O)_4(OH)_2] + 2H_2O$$

This is a deprotonation reaction – the two hydroxide ions have removed hydrogen ions from two of the water ligands and converted them into water molecules. The two water ligands that have lost hydrogen ions are now hydroxide ligands.

The same observations are made when aqueous ammonia is used as the alkali. However, when aqueous ammonia is added in excess, there is an observation not made with aqueous sodium hydroxide – the precipitate dissolves to form a deep blue solution. The equation for the deprotonation reaction forming the green precipitate is:

$$[Ni(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Ni(H_2O)_4(OH)_2] + 2NH_4^+$$

The equation for the ligand exchange reaction forming the blue solution is:

$$[Ni(H_2O)_4(OH)_2] + 6NH_3 \rightarrow [Ni(NH_3)_6]^{2+} + 4H_2O + 2OH^-$$

REACTIONS OF ZINC COMPLEXES WITH ALKALIS

Consider the reaction that occurs when aqueous sodium hydroxide is added to a solution containing the hexaaquazinc(II) ion until no further change is seen. The observation is that a colourless solution forms a white precipitate. The equation for the reaction is:

$$[Zn(H_2O)_6]^{2+} + 2OH^- \rightarrow [Zn(H_2O)_4(OH)_2] + 2H_2O$$

This is a deprotonation reaction – the two hydroxide ions have removed hydrogen ions from two of the water ligands and converted them into water molecules. The two water ligands that have lost hydrogen ions are now hydroxide ligands.

When an excess of aqueous sodium hydroxide is added, the white precipitate dissolves to form a colourless solution. The equation for this deprotonation reaction is:

$$[Zn(H_2O)_4(OH)_2] + 2OH^- \rightarrow [Zn(H_2O)_2(OH)_4]^{2-} + 2H_2O$$

Zinc hydroxide also reacts with acid, showing its amphoteric behaviour:

$$[Zn(H_2O)_4(OH)_2] + 2H^+ \rightarrow [Zn(H_2O)_6]^{2+}$$

Similar observations are made when aqueous ammonia is used as the alkali. The equation for the deprotonation reaction forming the white precipitate is:

$$[Zn(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Zn(H_2O)_4(OH)_2] + 2NH_4^+$$

The equation for the ligand exchange reaction forming the colourless solution is:

$$[Zn(H_2O)_4(OH)_2] + 4NH_3 \rightarrow [Zn(NH_3)_4]^{2+} + 4H_2O + 2OH^-$$

LEARNING TIP

Some texts quote the formula of the ammine complex as $[Zn(NH_3)_6]^{2+}$ rather than $[Zn(NH_3)_4]^{2+}$.

A research article in the 1993 edition of *Chemical Physics Letters* suggests that the coordination number can be either 4 or 6, but that $[Zn(NH_3)_4]^{2+}$ is favoured as it has a lower energy. Therefore, it is reasonable to say that both formulae are correct.

SUMMARY

Table A summarises the reactions of the hydrated d-block metal ions with aqueous sodium hydroxide.

ION IN SOLUTION	COLOUR OF SOLUTION	COLOUR OF PRECIPITATE	EFFECT OF ADDING EXCESS NaOH(aq)
$[Cr(H_2O)_6]^{3+}$	green	green	forms green solution
$[Mn(H_2O)_6]^{2+}$	pale pink	pale brown (turns darker brown on exposure to air)	none
$[Fe(H_2O)_6]^{2+}$	pale green	green (turns brown on exposure in air)	none
$[Fe(H_2O)_6]^{3+}$	yellow-brown	brown	none
$[Co(H_2O)_6]^{2+}$	pink	blue (turns pink on exposure to air)	none
$[Ni(H_2O)_6]^{2+}$	green	green	none
$[Cu(H_2O)_6]^{2+}$	blue	blue	none
$[Zn(H_2O)_6]^{2+}$	colourless	white	forms colourless solution

table A

Table B summarises the reactions of the hydrated d-block metal ions with aqueous ammonia.

ION IN SOLUTION	COLOUR OF SOLUTION	COLOUR OF PRECIPITATE	EFFECT OF ADDING EXCESS $NH_3(aq)$
$[Cr(H_2O)_6]^{3+}$	green	green	forms purple solution
$[Mn(H_2O)_6]^{2+}$	pale pink	pale brown (turns darker brown on exposure to air)	none
$[Fe(H_2O)_6]^{2+}$	pale green	green (turns brown on exposure in air)	none
$[Fe(H_2O)_6]^{3+}$	yellow-brown	brown	none
$[Co(H_2O)_6]^{2+}$	pink	blue (turns pink on exposure to air)	forms pale yellow solution (turns brown on exposure to air)
$[Ni(H_2O)_6]^{2+}$	green	green	forms deep blue solution
$[Cu(H_2O)_6]^{2+}$	blue	blue	forms deep blue solution
$[Zn(H_2O)_6]^{2+}$	colourless	white	forms colourless solution

table B

CHECKPOINT

1. The electronic configurations of nickel and zinc are:

Ni [Ar] $3d^8\ 4s^2$

Zn [Ar] $3d^{10}\ 4s^2$

(a) State why both elements are classified as d-block elements.

(b) State why nickel is classified as a transition metal, but zinc is not.

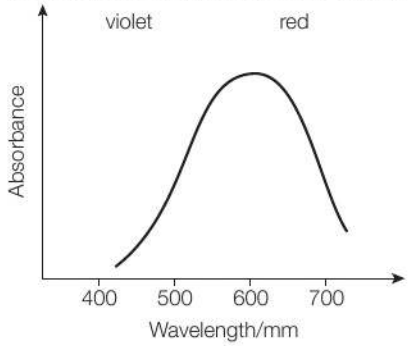
2. The EDTA4– ion forms a complex ion with the Ni2+ ion.

(a) Complete the electronic configuration of the Ni^{2+} ion.

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.....

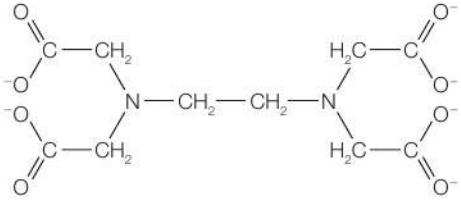
(b) The visible spectrum of the complex ion formed between Ni2+ and EDTA4– is shown below.

SKILLS ANALYSIS



Explain the colour of this complex ion.

(c) The $EDTA^{4-}$ ion has the following structure.



(i) Draw a ring around two different types of atoms in the $EDTA^{4-}$ ion that are capable of forming a dative (coordinate) bond with the Ni^{2+} ion.

(ii) What feature of these atoms allows them to form a bond with Ni^{2+}?